Generative AI Meets Knowledge Management: Insights From Software Development Practices

生成式 AI 與知識管理的交會：來自軟體開發實務的洞察

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▶️ **閱讀說明**　本版採「逐段中英對照」：每一段英文原文後緊接一個藍框中文譯文。本研讀聚焦核心章節，收錄範圍：摘要、第 1 節引言、第 2 節背景（2.1 KM 與 GenAI、2.2 GenAI 與軟體開發、2.3 TOE 框架）、第 3 節方法、第 4 節結果（4.1–4.5 五主題）、第 5 節討論、第 6 節結論。其中第 2.1.2 節密集回顧多篇他人實證研究，為節省篇幅以重點濃縮方式呈現並明註；表 2 之受訪者引述欄以重點濃縮處理，不逐字照錄訪談原話。

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摘要 Abstract

Recent developments in generative artificial intelligence (GenAI) have raised the interest of knowledge management (KM) scholars in artificial intelligence. By harnessing GenAI, KM processes become more efficient, scalable and adaptable to the needs of organisations and users. Taking the theoretical lens of the Technology-Organisation-Environment (TOE) framework for technology adoption, this study investigates the factors influencing the adoption of GenAI among software developers, as a specific group of knowledge workers, in knowledge creation and sharing. Our interviews with 11 developers from 8 countries were analysed by combining an inductive and a deductive approach. We identified five themes alongside the TOE framework, emphasising the adoption factors of GenAI in software development, particularly within KM.

> ▶ **中文譯文** 生成式人工智慧（GenAI）的近期發展，提高了知識管理（KM）學者對人工智慧的興趣。藉由運用 GenAI，知識管理流程變得更有效率、更可擴展，也更能因應組織與使用者的需求。本研究以技術組織環境（TOE）這套技術採用框架為理論視角，探討影響軟體開發者這一特定知識工作者群體在知識創造與分享中採用 GenAI 的各項因素。研究者對來自 8 個國家的 11 位開發者進行訪談，並結合歸納與演繹兩種取向加以分析。我們沿著 TOE 框架辨識出五個主題，凸顯軟體開發（特別是知識管理）情境中採用 GenAI 的因素。

Based on our findings, we discuss how GenAI is adopted for KM in software development. In particular, the interviewees liked the GenAI affordances of solving simpler programming tasks efficiently and rapidly. However, GenAI requires expertise to review and modify the code, write suitable prompts and evaluate the reliability of the provided output. Knowledge exchange with fellow programmers is partly, but not entirely, replaced by exchange with GenAI as a new development team member, as it is more efficient. Nevertheless, continuous learning, adaptation and ethical consideration are needed to realise the full benefits of GenAI tools. This study additionally prompts a critical reflection on the necessity of revising existing KM theoretical models in light of the emergence of AI-generated knowledge.

> ▶ **中文譯文** 基於研究發現，我們討論了 GenAI 如何被用於軟體開發的知識管理。特別是，受訪者喜歡 GenAI 能有效率且快速地解決較簡單程式任務的可供性（affordance）。然而，GenAI 需要專業能力來審閱與修改程式碼、撰寫合適的提示，並評估其所提供輸出的可靠性。與同行程式設計師的知識交流，部分但不完全地被「與 GenAI 這個新團隊成員的交流」所取代，因為後者更有效率。儘管如此，仍需要持續學習、適應與倫理考量，才能完整實現 GenAI 工具的效益。本研究並促使我們批判性地反思：面對 AI 生成知識的出現，是否有必要修訂既有的知識管理理論模型。

第一章　引言 1. Introduction

With advancements in technology, especially artificial intelligence (AI), the strategies and tools used in knowledge management (KM) are evolving. AI has various applications in KM, offering benefits such as real-time knowledge sharing, improving knowledge discovery, and enhancing knowledge delivery like adaptive learning and personalising knowledge distribution. According to Jarrahi et al. (2023), AI can support fundamental processes of KM, including the creation, storage, retrieval, sharing and application of knowledge. To sum up, it is a widespread opinion among scholars and practitioners that AI is offering new foundations to transform KM, as AI and KM at their core are about knowledge.

> ▶ **中文譯文** 隨著科技、特別是人工智慧（AI）的進展，知識管理（KM）所使用的策略與工具正在演變。AI 在知識管理中有多種應用，帶來如即時知識分享、改善知識探索，以及強化知識傳遞（如適性學習與個人化知識分配）等效益。根據 Jarrahi 等人（2023），AI 能支援知識管理的基本流程，包括知識

> 的創造、儲存、檢索、分享與應用。總而言之，學者與實務界普遍認為，AI 正為轉型知識管理提供新的基礎，因為 AI 與知識管理的核心都是「知識」。

Although AI has the potential to support KM in many ways, research on the relationships between AI and KM is still scarce, and there is little empirical evidence regarding integrating AI technologies into KM processes. Extant studies are primarily conceptual, based on the analysis of existing literature, and consider the entire set of AI technologies. Furthermore, they underline the advantages and benefits of AI while devoting less attention to the challenges and problems raised by its adoption in real-life cases.

> ▶️ **中文譯文**　儘管 AI 在許多面向都有支援知識管理的潛力，關於 AI 與知識管理之間關係的研究仍然稀少，將 AI 技術整合進知識管理流程的實證證據也很有限。現有研究大多偏向概念性，奠基於既有文獻的分析，並把整套 AI 技術一概而論。此外，這些研究多強調 AI 的優勢與效益，卻較少關注其在真實案例中採用時所引發的挑戰與問題。

It is a common opinion that the recent advances in AI, especially the public availability of generative AI (GenAI), such as GPT, Dall-E or Copilot, will change how we work and communicate. GenAI encompasses computational technologies that create new and meaningful content based on training data, such as text, images or audio. The development of the last version of ChatGPT, a very popular GenAI model, has raised great interest among scholars and practitioners regarding its possible use in organisational management and also in KM.

> ▶️ **中文譯文**　一個普遍的看法是，AI 的近期進展，尤其是 GPT、Dall-E 或 Copilot 等生成式 AI（GenAI）的公開可得，將改變我們工作與溝通的方式。GenAI 涵蓋一類能基於訓練資料創造出嶄新且有意義內容（如文字、影像或音訊）的運算技術。極受歡迎的 GenAI 模型 ChatGPT 最新版本的問世，在學者與實務界之間激起了高度興趣，關注其在組織管理乃至知識管理中可能的用途。

Our paper aims to provide insights into how AI can affect KM processes in knowledge-intensive organisations. In particular, it examines how software developers use GenAI to generate new knowledge, especially new software code, and share knowledge regarding development problems. Software developers are a specific group of knowledge workers who combine creativity in developing new applications (knowledge creation) and rational programming rules (explicit knowledge) with their expertise and experience (tacit knowledge). To address this research gap, we interviewed 11 software developers in eight countries about their experiences using GenAI daily and how it impacts knowledge creation and sharing. Furthermore, we employed the Technology-Organisation-Environment (TOE) framework of technology adoption as an additional theoretical lens to interpret the empirical findings.

> ▶️ **中文譯文**　本文旨在提供洞見，說明 AI 如何影響知識密集型組織中的知識管理流程。具體而言，本文檢視軟體開發者如何運用 GenAI 來生成新知識（特別是新的程式碼），並分享關於開發問題的知識。軟體開發者是一個特定的知識工作者群體，他們把開發新應用的創造力（知識創造）、理性的程式規則（外顯知識），與自身的專業與經驗（內隱知識）結合在一起。為了填補這個研究缺口，我們訪談了 8 個國家的 11 位軟體開發者，了解他們每日使用 GenAI 的經驗，以及它如何影響知識的創造與分享。此外，我們採用技術組織環境（TOE）這套技術採用框架，作為詮釋實證發現的額外理論視角。

第二章　研究背景 2. Background of the Study

2.1　知識管理與生成式 AI 2.1 KM and GenAI

2.1.1 知識管理與 AI 的交會 2.1.1 The Intersection of KM and AI

A strict link between KM and AI was highlighted more than two decades ago by Liebowitz (2001), who considered AI to be one of the critical blocks for developing and advancing the field of KM. In recent years, the opinion is rapidly diffused that the latest advances in AI cannot only provide new foundations for profoundly transforming KM in organisations but also that adopting a KM perspective can enable a better implementation of AI. Edwards and Lonnqvist (2023), in their paper on the future of KM, underline the need to go deeper into the connection between AI and KM, asserting that 'AI desperately needs KM' to exploit its opportunities fully.

> ▶️ **中文譯文** 早在二十多年前，Liebowitz（2001）就點出知識管理與 AI 之間的緊密連結，他認為 AI 是發展與推進知識管理領域的關鍵基石之一。近年來，一種看法迅速擴散：AI 的最新進展不僅能為深刻轉型組織中的知識管理提供新基礎，採取知識管理的視角也能促成 AI 更好的落實。Edwards 與 Lönnqvist（2023）在其探討知識管理未來的論文中，強調需要更深入地探究 AI 與知識管理之間的連結，並主張「AI 迫切需要知識管理」，才能充分發揮其機會。

GenAI uses advanced algorithms called large language models (LLMs) based on deep learning and artificial neural networks to create new content like text, images and media. GenAI can process natural language to help employees retrieve stored internal knowledge by phrasing questions like they might ask a human and engage in an ongoing dialogue. All the previous features, particularly its back-and-forth conversational mechanism, make GenAI the most considerable disruption in the KM field.

> ▶️ **中文譯文** GenAI 運用一種稱為大型語言模型（LLM）的先進演算法，這類模型奠基於深度學習與人工神經網路，能創造出文字、影像與媒體等新內容。GenAI 能處理自然語言，讓員工以「彷彿在問人」的方式提問來檢索已儲存的內部知識，並進行持續的對話。前述所有特性，尤其是其一來一往的對話機制，使 GenAI 成為知識管理領域中最具份量的顛覆性力量。

Alavi et al. (2024) analysed the organisational implications of GenAI from a KM perspective. They investigated how GenAI impacts the different KM processes (creation, storage, transfer and application). Regarding knowledge creation, the authors affirm that GenAI can facilitate the combination and internalisation modes of knowledge creation. As regards the combination mode, GenAI excels at generating new content, identifying patterns and making predictions, serving as an active knowledge co-creator. Furthermore, GenAI supports internalisation through interactive, conversational engagement of users.

> ▶️ **中文譯文** Alavi 等人（2024）從知識管理視角分析了 GenAI 的組織意涵。他們探討 GenAI 如何影響不同的知識管理流程（創造、儲存、轉移與應用）。在知識創造方面，作者指出 GenAI 能促進知識創造中的「結合」（combination）與「內化」（internalisation）模式。就結合模式而言，GenAI 擅長生成新內容、辨識模式並做出預測，扮演主動的知識共創者。此外，GenAI 透過與使用者的互動式對話來支援內化。

Together with the opportunities, Alavi et al. (2024) identified six critical areas: (1) a loss of control over the knowledge generated by GenAI; (2) the risk that intense resort to GenAI may reduce human socialisation and knowledge collaboration; (3) the sidestepping of junior knowledge workers, whose entry tasks are now performed by GenAI; (4) the lowering of knowledge internalisation, as employees may accept GenAI knowledge without internalising it; (5) the possible dissuasion of employees from engaging in the externalisation of knowledge; and (6) the possible generation of inaccurate outputs (the so-called 'hallucinations').

> ▶️ **中文譯文** 在機會之外，Alavi 等人（2024）辨識出六個關鍵風險領域：（1）對 GenAI 所生成知識失去掌控；（2）過度倚賴 GenAI 可能減少人際社會化與知識協作的風險；（3）初階知識工作者被邊

> 緣化，因為他們的入門任務如今由 GenAI 執行；（4）知識內化的降低，因為員工可能接受 GenAI 生成的知識卻未加以內化；（5）員工可能因此不再投入於知識的外化（externalisation）；以及（6）可能產生不正確的輸出（即所謂的「幻覺」）。

> （研讀範圍說明）原文第 2.1.2 節「KM 與 GenAI 整合的實證研究」逐一回顧 Woodruff 等（2024）、Brynjolfsson 等（2023）、Ritala 等（2024）、Kaczorowska-Spychalska 等（2024）、Sumbal 等（2024）、Retkowsky 等（2024）等多篇研究。重點濃縮如下：這些早期實證研究數量仍少，共同指向三項要點——GenAI 對知識管理的影響具高度情境依賴（隨產業、角色、認知工作性質而異）；GenAI 兼具強化與阻礙組織知識管理的雙面效果；其利弊取決於所支援的特定知識管理活動。作者因此主張，研究應從通論轉向特定應用情境的深掘，這也是本研究選擇「軟體開發」作為切入場域的理由。

2.2 生成式 AI 與軟體開發 2.2 GenAI and Software Development

The adoption of GenAI affects software development by offering new opportunities, methodologies, and tools that significantly boost efficiency, minimise manual effort and foster creative programming practices. GenAI has demonstrated its ability to enhance productivity, reduce human errors, automate code generation and documentation and find errors in software code. By streamlining these processes, GenAI can improve the quality of work and allow developers to focus on more complex, knowledge-intensive and creative aspects of their projects.

> ▶️ **中文譯文** GenAI 的採用，藉由提供新的機會、方法與工具，顯著提升效率、減少人工投入並培養具創造力的程式實務，從而影響軟體開發。GenAI 已展現其能力，能提升生產力、減少人為錯誤、自動化程式碼生成與文件撰寫，並找出程式碼中的錯誤。藉由簡化這些流程，GenAI 能改善工作品質，讓開發者把心力集中在專案中更複雜、更知識密集且更具創造性的部分。

Most studies show that GenAI performs well on small, well-defined coding tasks and can produce high-quality code, but struggles with larger, more complex problems. Human oversight remains crucial to validate accuracy and ensure the code meets project requirements, highlighting the need for a balance between AI assistance and human judgement in the software development process. GenAI also plays a critical role in collaborative development; however, developers increasingly rely on AI for solutions instead of colleagues, which can disrupt traditional knowledge-sharing practices within agile teams.

> ▶️ **中文譯文** 多數研究顯示，GenAI 在小型、定義明確的程式任務上表現良好，能產出高品質的程式碼，但在較大、較複雜的問題上則力有未逮。人工監督仍然至關重要，用以驗證正確性並確保程式碼符合專案需求，這凸顯了在軟體開發過程中，AI 協助與人類判斷之間取得平衡的必要。GenAI 在協作開發中也扮演關鍵角色；然而，開發者愈來愈倚賴 AI 而非同事來取得解答，這可能擾亂敏捷團隊中傳統的知識分享實務。

Furthermore, Komp-Leukkunen (2024) discusses how tools like ChatGPT could democratise programming knowledge, enabling individuals with little formal education to acquire programming skills independently. This democratisation, while expanding the talent pool, may reduce the specialist status of software engineers. One effective strategy is leveraging GenAI to review tasks rather than generate design documents or code, as mistakes in review outputs tend to be less severe than errors in generated code. To sum up, GenAI is rapidly transforming the software development industry, which has been among the first and most intense adopters of such technology. What currently needs to be added, however, is an analysis based on a KM perspective.

> ▶️ **中文譯文** 此外，Komp-Leukkunen（2024）討論了 ChatGPT 這類工具如何能使程式知識民主化，讓正式教育不多的人也能獨立習得程式技能。這種民主化雖擴大了人才庫，卻可能降低軟體工程師的

專家地位。一個有效的策略是運用 GenAI 來「審查」任務，而非生成設計文件或程式碼，因為審查輸出中的錯誤，其嚴重性通常低於生成程式碼中的錯誤。總而言之，GenAI 正迅速轉變軟體開發產業，而該產業正是這類技術最早、也最積極的採用者之一。然而，目前仍需補上的，是一個奠基於知識管理視角的分析。

2.3　以 TOE 框架視角分析 GenAI 對知識管理的影響 2.3 Analyzing the Impact of GenAI on KM Through the TOE Framework Lens

Among the various theories and frameworks used to study the adoption of new technologies, we have chosen to employ the TOE framework developed by Tornatzky et al. (1990). This choice is based on the framework's previous practical application in studying the adoption of AI in general and of GenAI, and it was already used in the field of KM. Moreover, this framework can be employed to investigate and analyse various factors due to its inherent malleability and flexibility, making it well-suited for exploration without being constrained by predetermined constructs.

▶️ **中文譯文**　在研究新技術採用所使用的各種理論與框架中，我們選擇採用 Tornatzky 等人（1990）發展的 TOE 框架。此一選擇，是基於該框架先前在研究 AI 乃至 GenAI 採用上的實務應用，且它已被用於知識管理領域。此外，由於其本身具有可塑性與彈性，這套框架能用來探究與分析各種因素，使其非常適合在不受預設構念限制的情況下進行探索。

The TOE framework posits that three main factors influence an organisation's technology adoption decisions: (1) Technological factors refer to the characteristics of the new technology, including perceived benefits, compatibility, challenges and infrastructure, often traced back to technology affordances and constraints. (2) Organisational factors include the characteristics and resources of the internal organisational context, such as size, managerial structure and working procedures. (3) Environmental factors encompass the external environment, including industry requirements, competitive pressures, the legal and regulatory environment and ethical considerations.

▶️ **中文譯文**　TOE 框架主張，有三大因素影響組織的技術採用決策：（1）技術因素，指新技術的特性，包括感知效益、相容性、挑戰與基礎設施，常可追溯至技術可供性（affordances）與技術限制（constraints）；（2）組織因素，包含內部組織情境的特性與資源，如規模、管理結構與工作程序；（3）環境因素，涵蓋組織所處的外部環境，包括產業要求、競爭壓力、法律與監管環境，以及倫理考量。

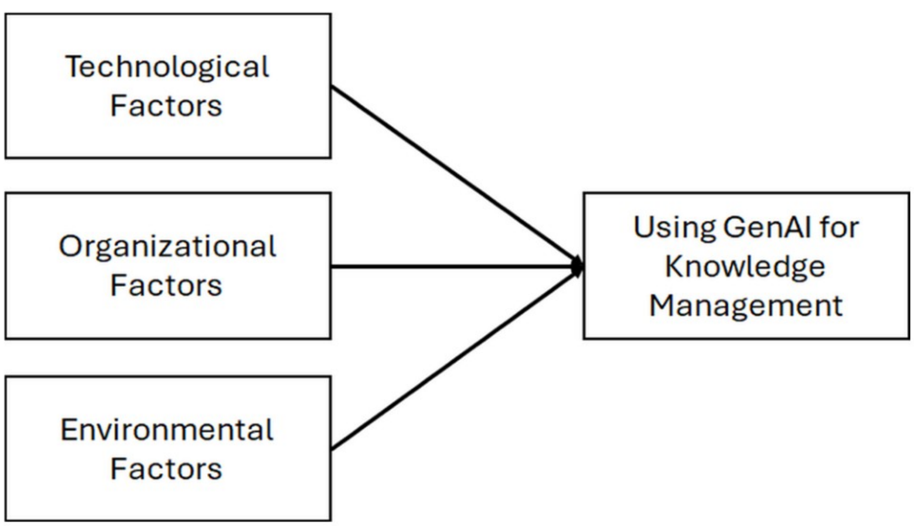

Figure 1. Applying the TOE framework to investigate the GenAI influence on knowledge management.

圖 1　以 TOE 框架探究 GenAI 對知識管理的影響：技術、組織、環境三因素共同作用於「運用 GenAI 於知識管理」

第三章　研究方法 3. Research Methodology

3.1　研究設計 3.1 Research Design

This study investigated the purpose and use of GenAI during software development, which is considered a knowledge-intensive activity. It examined the main benefits and challenges connected with its use, the new competencies required for software developers using GenAI, and the impact it may have on interactions with colleagues and onboarding new hires. The findings were then interpreted from a KM perspective, focusing on the knowledge generation and sharing processes inherent in software development tasks. Then, we applied the TOE framework for technology adoption as a theoretical lens, with GenAI as a technology that influences the KM processes related to the software development activity.

> ▶ **中文譯文**　本研究探討 GenAI 在軟體開發過程中的目的與使用，而軟體開發被視為一種知識密集型的活動。研究檢視了使用 GenAI 所連帶的主要效益與挑戰、軟體開發者使用 GenAI 所需的新能力，以及它對「與同事互動」及「新進人員到職培訓」可能造成的影響。研究發現隨後從知識管理的視角加以詮釋，聚焦於軟體開發任務中固有的知識生成與分享流程。接著，我們以 TOE 技術採用框架作為理論視角，將 GenAI 視為一項影響軟體開發活動相關知識管理流程的技術。

Given the exploratory aim of the research, we adopted a qualitative approach, applying a convenience sampling method. During February 2024, we conducted 11 semi-structured interviews with software developers from eight countries and with different levels of programming experience, aiming to capture a broad spectrum of experiences and insights. The interviews were audio-recorded and transcribed verbatim. We combined an inductive and a deductive approach: first following Braun and Clarke's (2006) six-phase framework to let themes emerge directly from interviews, then applying the TOE framework as a theoretical lens to interpret and organise the findings.

> ▶ **中文譯文**　基於研究的探索性目的，我們採用質化取向，運用便利抽樣法。在 2024 年 2 月，我們對來自 8 個國家、具不同程度程式經驗的軟體開發者進行了 11 場半結構式訪談，以期捕捉廣泛的經驗與洞見。訪談經錄音並逐字轉錄。我們結合歸納與演繹兩種取向：先依循 Braun 與 Clarke（2006）的六階段框架，讓主題直接從訪談中浮現；再以 TOE 框架作為理論視角，來詮釋並組織這些發現。

3.2　受訪者輪廓 3.2 Profile of Interview Respondents

Table 1 summarises the profile of respondents, whose diverse experiences and backgrounds in software development and GenAI tools enrich the understanding of the impact of GenAI in different contexts. The interviewees span from self-employment and startups to employees at large corporations in the technology, finance, security and gaming sectors, showcasing wide geographical and cultural diversity. They hold various roles, including technical leadership, coding and academic research, with experiences ranging from short to very long.

> ▶ **中文譯文**　表 1 摘要了受訪者的輪廓，他們在軟體開發與 GenAI 工具上多元的經驗與背景，豐富了我們對 GenAI 在不同情境下影響的理解。受訪者橫跨自雇者與新創公司，到科技、金融、資安與遊戲產業大型企業的員工，展現出廣泛的地理與文化多樣性。他們擔任各種角色，包括技術領導、程式撰寫與學術研究，年資從短到非常長皆有。

表 1　受訪者輪廓（共 11 位） *Table 1. Profile of respondents (n = 11).*

編號 No	國家 Country	角色 Role	軟體年資 Yrs Dev	GenAI 年資／產業
1	丹麥	資深開發者	30	1 年｜軟體公司
2	義大利	全端開發者	2	1 年｜顧問公司
3	義大利	新創負責人／開發者	4	1.5 年｜新創、IT 服務
4	丹麥	初階開發者	6	1 年｜研究機構
5	德國	初階開發者	0.5	0.5 年｜新聞通訊社
6	伊朗	技術團隊領導	8	2 年｜金融服務
7	英國	軟體工程師	3	0.5 年｜軟體公司
8	瑞典	首席軟體工程師	10	0 年｜遊戲產業
9	伊朗	中階開發者	14	2 年｜金融服務
10	法國	資深開發者	28	0 年｜軟體公司
11	荷蘭	資深開發者	25	1 年｜軟體公司

第四章　結果 4. Results

We employed a structured clustering approach to analyse the interview responses, identifying five key themes: (1) usage and purposes for GenAI; (2) challenges with GenAI; (3) learning and adaptation; (4) current work practices integration; (5) public data and privacy. Following the TOE framework, the first two pertain to technological aspects, the next two to organisational aspects, and the final one to environmental aspects. Therefore, the empirical findings highlight that all three factors of the TOE framework are relevant in understanding how software developers adopt GenAI tools to support their KM-related tasks.

> ▶️ **中文譯文** 我們採用結構化分群法來分析訪談回應，辨識出五個關鍵主題：（1）GenAI 的使用與目的；（2）使用 GenAI 的挑戰；（3）學習與適應；（4）現行工作實務的整合；（5）公開資料與隱私。依循 TOE 框架，前兩者屬於技術面向，接下來兩者屬於組織面向，最後一者屬於環境面向。因此，實證發現凸顯出：TOE 框架的三個因素，對於理解軟體開發者如何採用 GenAI 工具來支援其知識管理相關任務，都具有相關性。

4.1　技術可供性：使用與目的 4.1 Technological Affordances: Usage and Purposes

The interviewed software developers increasingly rely on GenAI tools for various knowledge-intensive tasks, including debugging, code generation, code optimisation, and code refactoring. This confirms the malleable nature of GenAI technology. Its adoption aims to enhance the speed and quality of the development process, directly impacting KM by accelerating knowledge creation and application within projects. Some interviewees indicated that they use the tool to handle unfamiliar algorithms and for refactoring code. Others noted that by precisely describing their problem, they can receive a proposed solution. This shift underscores a move towards more efficient and automated KM, where knowledge is actively generated and applied in real-time to solve complex problems.

> ▶️ **中文譯文** 受訪的軟體開發者愈來愈倚賴 GenAI 工具來處理各種知識密集型任務，包括除錯、程式碼生成、程式碼優化與程式碼重構。這印證了 GenAI 技術的可塑性。採用它的目的在於提升開發過程的速度與品質，並透過加速專案內的知識創造與應用，直接影響知識管理。部分受訪者表示，他們用這

項工具來處理不熟悉的演算法，以及重構程式碼。另一些人則指出，只要精確描述問題，就能得到建議的解法。這種轉變凸顯出朝向更有效率、更自動化知識管理的移動，知識在其中被即時主動生成並應用，以解決複雜問題。

4.2 技術限制：使用 GenAI 的挑戰 4.2 Technological Constraints: Challenges

Developers may encounter problems such as code inaccuracies, a lack of context understanding by the GenAI, and difficulties debugging GenAI-generated code. There are concerns about the reliability and security of the code produced, as well as potential biases inherited from the training data. There is also the risk of over-reliance on GenAI tools, which might hinder the development of developers' coding skills and critical thinking abilities. A respondent expressed that they do not fully trust the tool and feel the need to verify its outputs, using it as an initial proposal which they then review and often need to modify.

▶ **中文譯文** 開發者可能遭遇諸如程式碼不準確、GenAI 缺乏脈絡理解，以及難以對 GenAI 生成的程式碼除錯等問題。對於所產出程式碼的可靠性與安全性，以及可能從訓練資料繼承而來的偏誤，都存在疑慮。此外也有過度倚賴 GenAI 工具的風險，這可能妨礙開發者程式技能與批判思考能力的養成。一位受訪者表示，他並不完全信任這項工具，覺得有必要驗證它的輸出，把它當作初步的提案，再加以審閱，且往往需要修改。

GenAI significantly influences the knowledge sources utilised by software developers. In particular, GenAI reduces the reliance on traditional sources such as forums (e.g., Stack Overflow), blogs and consultations with colleagues. This shift is primarily due to GenAI's ability to offer rapid and straightforward solutions, eliminate the need for consulting multiple sources, and facilitate the interactive reformulation of questions. However, despite these advantages, GenAI is considered less reliable than traditional knowledge sources, especially for addressing uncommon, very specific or new problems. Reviewing the code provided as a solution requires a high level of expertise to determine whether it is correct.

▶ **中文譯文** GenAI 顯著影響了軟體開發者所運用的知識來源。具體而言，GenAI 降低了對論壇（如 Stack Overflow）、部落格與同事諮詢等傳統來源的依賴。這種轉變主要源於 GenAI 能提供快速而直接的解法、免去查閱多個來源的需要，並支援以互動方式重新表述問題。然而，儘管有這些優勢，GenAI 仍被認為比傳統知識來源不可靠，尤其在處理罕見、非常特定或全新的問題時。審閱作為解法提供的程式碼，需要高度的專業才能判斷其是否正確。

4.3 組織因素：學習與適應 4.3 Organisational Factors: Learning and Adaptation

GenAI requires users to ask accurate questions and clearly define problems, necessitating new competencies in question formulation. This necessity has led to the emergence of a new discipline known as prompt engineering. As developers engage with GenAI tools, they acquire new technical skills and develop sophisticated problem-solving abilities. A critical question that emerged is whether GenAI can be considered a 'new' team member. Some respondents noted that using GenAI leads to fewer interruptions when asking colleagues for help. However, this also results in less interaction between colleagues, reducing opportunities for one question to spark another and potentially limiting the creation and sharing of new knowledge.

▶ **中文譯文** GenAI 要求使用者提出準確的問題並清楚界定問題，這需要在「問題構築」上具備新的能力。這種必要性催生了一個稱為「提示工程」（prompt engineering）的新學門。當開發者投入使用 GenAI 工具，他們習得新的技術技能，並發展出精細的問題解決能力。一個浮現的關鍵問題是：GenAI 是否可被視為一個「新的」團隊成員。部分受訪者指出，使用 GenAI 使得向同事求助的打擾次

數減少。然而，這也導致同事之間的互動變少，減少了「一個問題激發另一個問題」的機會，並可能限制新知識的創造與分享。

4.4　組織因素：現行工作實務的整合 4.4 Organisational Factors: Current Work Practices Integration

GenAI tools are integral in creating a knowledge-rich environment by improving efficiency, facilitating collaboration and enhancing product quality. They enable the seamless integration of new hires into projects, effectively managing tacit and explicit knowledge within teams. As derived from the interviews, one of the significant skills sought when hiring people is general knowledge and the ability to utilise GenAI. In the future, soft skills and creativity will be crucial, especially in software development, where the ability to view issues from different angles, essential for generating new knowledge, will be vital.

▶️ **中文譯文**　GenAI 工具藉由提升效率、促進協作與強化產品品質，是打造知識豐富環境不可或缺的一環。它們讓新進人員得以無縫整合進專案，有效管理團隊內的內隱與外顯知識。如訪談所得，招募人員時看重的重要技能之一，是具備通用知識以及運用 GenAI 的能力。展望未來，軟性技能與創造力將至關重要，尤其在軟體開發中，能從不同角度看待問題的能力——這對生成新知識至關重要——將格外關鍵。

4.5　環境因素：公開資料與隱私 4.5 Environmental Factors: Public Data and Privacy

This theme concerns the data sources that GenAI tools rely on, and data privacy and security concerns, especially when proprietary codebases might be exposed to external AI services. Developers and organisations are wary of the potential for sensitive information leakage and the implications for intellectual property rights. Some respondents stated that their companies are not allowing them to use these kinds of tools with sensitive company data due to company policies and legal issues. However, openly sharing code with ChatGPT was viewed positively when possible, provided the code is not confidential or proprietary (e.g., for research or open-source development).

▶️ **中文譯文**　本主題關注 GenAI 工具所仰賴的資料來源，以及資料隱私與安全的疑慮，尤其當專屬程式碼庫可能暴露給外部 AI 服務時。開發者與組織對敏感資訊外洩的可能性，以及對智慧財產權的影響有所戒慎。部分受訪者表示，基於公司政策與法律問題，公司不允許他們以這類工具處理敏感的公司資料。然而，在可行的情況下，只要程式碼並非機密或專屬（例如用於研究或開源開發），公開地把程式碼分享給 ChatGPT 則被正面看待。

表 2　以知識管理視角綜整研究發現（受訪引述已重點濃縮） *Table 2. Summary of findings in light of knowledge management (interviewee quotes condensed).*

主題 Theme	TOE	洞見 Insights
GenAI 使用與目的	技術	自動化除錯與程式式生成以提升效率，加速開發；知識管理焦點轉向即時的知識生成與應用
GenAI 的挑戰	技術	需要穩健的知識管理實務來驗證與整合 AI 生成知識；不加批判地使用會妨礙真正新知識的產生
學習與適應	組織	培養動態學習環境，強化技能；但也可能減少同儕互動，限制自發的知識分享與協作學習
現行工作實務整合	組織	轉變開發實務，打造知識豐富環境；強化新進人員到職與專案整合，活用內隱與外顯知識

主題 Theme	TOE	洞見 Insights
公開資料與隱私	環境	引發知識來源與運用的疑慮，需透明的知識管理政策以維護資料完整性與機密性

第五章　討論 5. Discussion

Our study offers empirical evidence on the impact of GenAI on software development practices, which we examined through the double lens of the TOE framework and KM notions. Similar to other findings, our empirical study found that GenAI is helpful in many aspects of software development, especially in code generation, software testing and learning. Respondents were, however, also aware of the challenges that GenAI use brings. Our interview partners experienced difficulties due to errors in the GenAI-generated knowledge (the software code). Furthermore, they revealed that they rely more on GenAI to get help with their coding issues than to reach out to colleagues, as GenAI is simply quicker in providing solutions.

> ▶️ **中文譯文**　本研究提供了關於 GenAI 對軟體開發實務影響的實證證據，我們透過 TOE 框架與知識管理概念的雙重視角加以檢視。與其他研究的發現相似，我們的實證研究發現 GenAI 在軟體開發的許多面向都有幫助，特別是在程式碼生成、軟體測試與學習方面。然而，受訪者也意識到使用 GenAI 所帶來的挑戰。我們的訪談對象經歷了因 GenAI 生成知識（即程式碼）出錯而產生的困難。此外，他們透露，比起向同事求助，他們更倚賴 GenAI 來協助處理程式問題，因為 GenAI 提供解法就是比較快。

As Alavi et al. (2024) discussed, there is a risk that knowledge workers reduce their collaboration and exchange with other knowledge workers because GenAI takes this role. Although we can see this effect in our data, the investigated developers still reach out to human colleagues when the development problem is very specific, so GenAI can only help a little. In contrast to Alavi et al. (2024), our study revealed a more nuanced view on junior workers: while fewer young developers may be needed as GenAI overtakes simple tasks, GenAI can also help onboard and train young developers, who still need their general development knowledge to understand the GenAI-generated code.

> ▶️ **中文譯文**　如 Alavi 等人（2024）所討論，存在一種風險：知識工作者會因為 GenAI 取代了這個角色，而減少與其他知識工作者的協作與交流。儘管我們在資料中能看到這種效應，但所研究的開發者在開發問題非常特定時，仍會求助於人類同事，此時 GenAI 只能幫上一點忙。與 Alavi 等人（2024）不同，我們的研究對初階工作者呈現出更細緻的觀點：雖然當 GenAI 接手簡單任務後，所需的年輕開發者可能變少，但 GenAI 也能協助年輕開發者到職與受訓，而他們仍需要自身的通用開發知識，才能理解 GenAI 生成的程式碼。

Even if focused on software development, our analysis confirms that GenAI is assuming the role of a game changer in organisational KM: KM approaches and activities are required to change to seize the opportunities and face the challenges given by GenAI. For instance, the fact that the artificially generated knowledge may be irrelevant, if not inaccurate, calls for its continuous verification. This requires not completely abandoning the traditional knowledge sources, favouring the development of a critical mind, and avoiding a passive use of the artificially generated knowledge. To counteract the possible decrease of knowledge sharing among colleagues, the organisation should rethink socialisation and knowledge-sharing practices.

> ▶️ **中文譯文**　即便聚焦於軟體開發，我們的分析仍確認 GenAI 正在組織知識管理中扮演「改變遊戲規則者」的角色：知識管理的取向與活動必須改變，才能掌握 GenAI 帶來的機會、面對其挑戰。舉例而言，人工生成的知識可能不相關、甚至不準確，這就要求對其進行持續的驗證。這需要不完全捨棄傳統知

識來源、鼓勵批判心智的養成，並避免被動地使用人工生成的知識。為了抵銷同事間知識分享可能的減少，組織應重新思考社會化與知識分享的實務做法。

第六章　結論 6. Conclusion

The present study aimed to deepen our understanding of the impact of GenAI on organisational KM, specifically on knowledge creation and knowledge-sharing processes. We conducted 11 interviews with software developers who used GenAI in their daily activities. The study's findings shed light on the current usage, benefits and challenges of using GenAI for software code generation and the generation of new knowledge. In particular, the interviews revealed that GenAI can help developers solve simpler, repetitive tasks, saving them much time. However, the source for the generated code is unclear, and developers need to learn how to formulate their questions to the GenAI tool to receive good results.

▶ **中文譯文** 本研究旨在深化我們對 GenAI 影響組織知識管理的理解，特別是在知識創造與知識分享流程方面。我們訪談了 11 位在日常工作中使用 GenAI 的軟體開發者。研究發現闡明了當前使用 GenAI 進行程式碼生成與新知識生成的用途、效益與挑戰。具體而言，訪談揭示 GenAI 能幫助開發者解決較簡單、重複性的任務，為他們節省大量時間。然而，生成程式碼的來源並不清楚，開發者需要學習如何向 GenAI 工具構築自己的問題，才能得到良好的結果。

From an academic perspective, this study expands the currently limited empirical evidence on using GenAI to support KM processes. Furthermore, as we find that AI-generated knowledge represents a new type of knowledge that impacts organisational KM, we recommend that future work investigates this knowledge type further, particularly examining how existing KM frameworks may need to be revised to integrate AI-generated knowledge. From the managerial point of view, as these technologies continue to evolve, so must the KM practices that support their adoption, ensuring that knowledge is not only efficiently created and shared but also ethically managed and protected.

▶ **中文譯文** 從學術視角來看，本研究擴充了目前仍有限的、關於運用 GenAI 支援知識管理流程的實證證據。此外，由於我們發現 AI 生成知識代表一種影響組織知識管理的新型知識，我們建議未來研究進一步探討這種知識類型，特別是檢視既有的知識管理框架可能需要如何修訂，才能整合 AI 生成知識。從管理視角來看，隨著這些技術持續演進，支援其採用的知識管理實務也必須隨之演進，以確保知識不僅被有效率地創造與分享，也能被合乎倫理地管理與保護。

The interviews provide specific practical implications: (a) Re-focus KM activities by considering the type of knowledge GenAI can manage, requiring AI literacy and recognising prompt engineering as a new form of knowledge creation; (b) Adopt validation procedures where human experts tag and review outputs before integration into knowledge repositories; (c) Encourage active knowledge exchanges between team members, combining AI-generated knowledge with human team input to balance speed and richness; (d) Enhance onboarding by integrating GenAI alongside traditional training, recognising the ability to validate AI outputs and design prompts as a new and necessary skill set.

▶ **中文譯文** 訪談提供了具體的實務意涵：（a）重新聚焦知識管理活動，考量 GenAI 能管理的知識類型，這需要 AI 素養，並把提示工程視為一種新的知識創造形式；（b）採行驗證程序，由人類專家在輸出整合進知識庫之前先標記並審閱；（c）鼓勵團隊成員間主動的知識交流，把 AI 生成知識與人類團隊投入相結合，以在速度與豐富度之間取得平衡；（d）強化到職培訓，把 GenAI 與傳統訓練方式整合，並將「驗證 AI 輸出與設計提示的能力」視為一種新的、必要的技能組合。

Our study faces limitations due to the early adoption phase of GenAI among software developers. Currently, GenAI's application is mainly personal, with minimal integration into team processes, reflecting its limited influence on collective

KM practices at this stage. Another limitation stems from the reliance on interviewees' perceptions, which we attempted to mitigate by including workers from diverse countries, roles and seniority levels. Regarding future research, the findings should first be subjected to empirical confirmation, for example through surveys utilising the TOE model constructs, integrated with the factors identified in this analysis.

> ▶️ **中文譯文** 本研究受限於 GenAI 在軟體開發者間仍處於早期採用階段。目前，GenAI 的應用主要是個人層次的，鮮少整合進團隊流程，反映出現階段它對集體知識管理實務的影響仍有限。另一項限制來自對受訪者主觀認知的依賴，我們試圖透過納入來自不同國家、角色與年資的工作者來加以緩解。在未來研究方面，這些發現應先接受實證確認，例如透過運用 TOE 模型構念、並整合本分析所辨識因素的問卷調查。